

Lab 3: Lines, Vectors and Orthogonality in $\mathbb{R}^2$ Solutions

Math 124, Fall 2026 at the University of Michigan

due by the end of class on Wednesday, September 16, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, most of which will involve writing math on paper, and some of which will involve running code in a Jupyter Notebook. Lab activities are meant to last an hour, and the second hour of lab is dedicated to starting the homework assignment. To receive credit for a lab, you must show your lab TA your work on both the lab worksheet and homework assignment.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Activity 1: Homogeneous and Inhomogeneous Equations

Classify each linear equation as *homogeneous* or *inhomogeneous*.

a) $3x - 4y = 0$

☐ Homogeneous ☐ Inhomogeneous

Solution: ☒ Homogeneous ☐ Inhomogeneous

A linear equation in two variables is homogeneous if it can be written in the form

$$ax + by = 0.$$

Otherwise, it is inhomogeneous.

b) $x + 2y = 5$

☐ Homogeneous ☐ Inhomogeneous

Solution: ☐ Homogeneous ☒ Inhomogeneous

c) $2x - y = 0$

☐ Homogeneous ☐ Inhomogeneous

Solution: ☒ Homogeneous ☐ Inhomogeneous

d) $x - 4y + 7 = 0$

☐ Homogeneous ☐ Inhomogeneous

Solution: ☐ Homogeneous ☒ Inhomogeneous

Activity 2: The Span of One Vector

Let

$$\vec{v} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

- a) Write $\text{span}(\vec{v})$ explicitly in set-builder notation.

$$\text{span}(\vec{v}) = \left\{ \right. \left. \right\}.$$

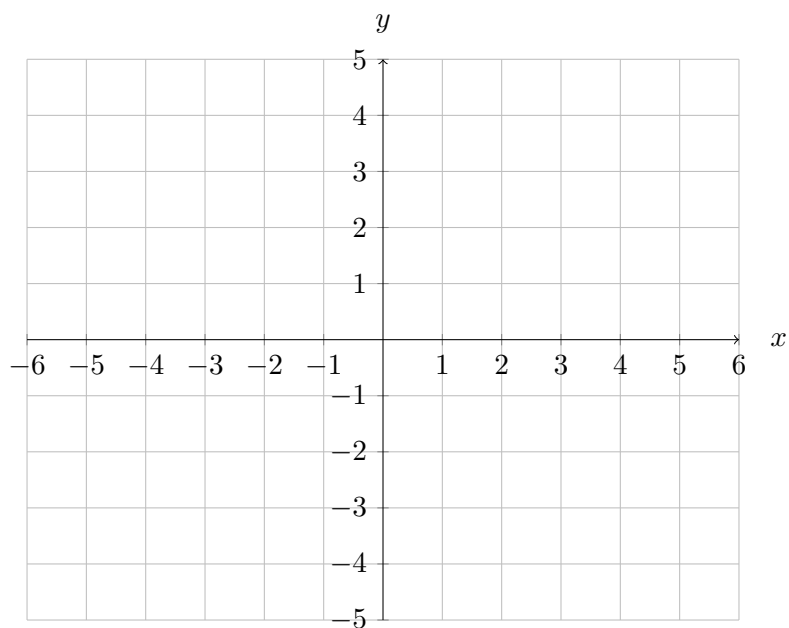
Solution:

$$\text{span}(\vec{v}) = \left\{ t \begin{bmatrix} 2 \\ -3 \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Equivalently,

$$\text{span}(\vec{v}) = \left\{ \begin{bmatrix} 2t \\ -3t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

- b) Draw $\text{span}(\vec{v})$ on the coordinate plane below. Label the line ℓ , and mark the vector $\vec{v}$.

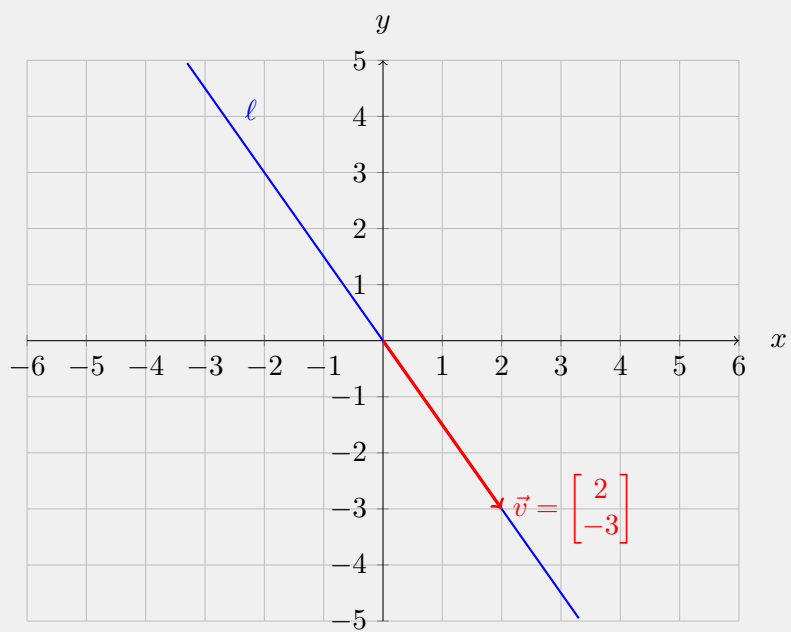


Solution: The graph is the line through the origin with direction vector

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Its equation is

$$3x + 2y = 0.$$



Activity 3: Translating a Line

Let

$$\vec{v}_0 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad \text{and} \quad \vec{v} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

a) Write

$$\vec{v}_0 + \text{span}(\vec{v})$$

explicitly in set-builder notation.

$$\vec{v}_0 + \text{span}(\vec{v}) = \left\{ \right. \left. \right\}.$$

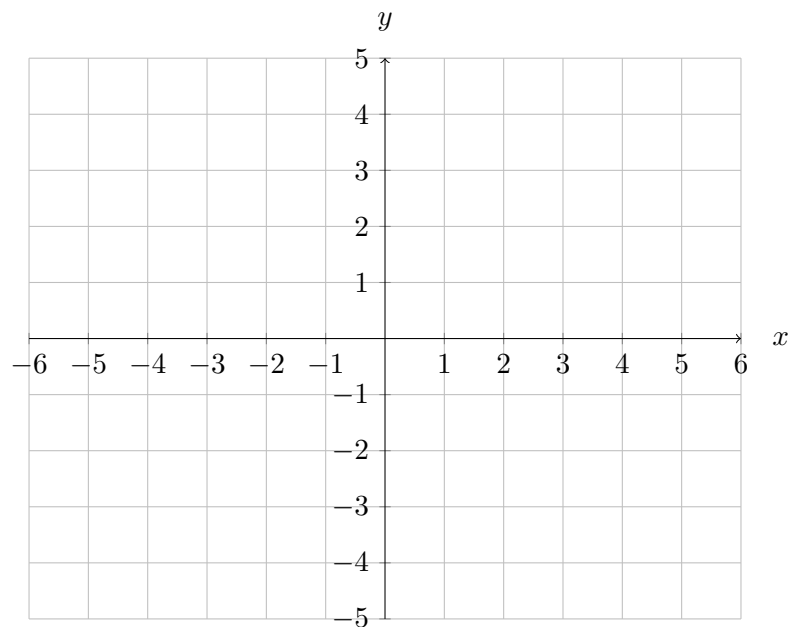
Solution:

$$\vec{v}_0 + \text{span}(\vec{v}) = \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} + t \begin{bmatrix} 2 \\ -3 \end{bmatrix} : t \in \mathbb{R} \right\}.$$

Equivalently,

$$\vec{v}_0 + \text{span}(\vec{v}) = \left\{ \begin{bmatrix} -2 + 2t \\ 1 - 3t \end{bmatrix} : t \in \mathbb{R} \right\}.$$

b) Draw this set on the coordinate plane below. Label the affine line ℓ' and mark the points represented by $\vec{v}_0$, $\vec{v}_0 + \vec{v}$, $\vec{v}_0 - \vec{v}$.



Solution: The three requested points are

$$\vec{v}_0 = \begin{bmatrix} -2 \\ 1 \end{bmatrix},$$

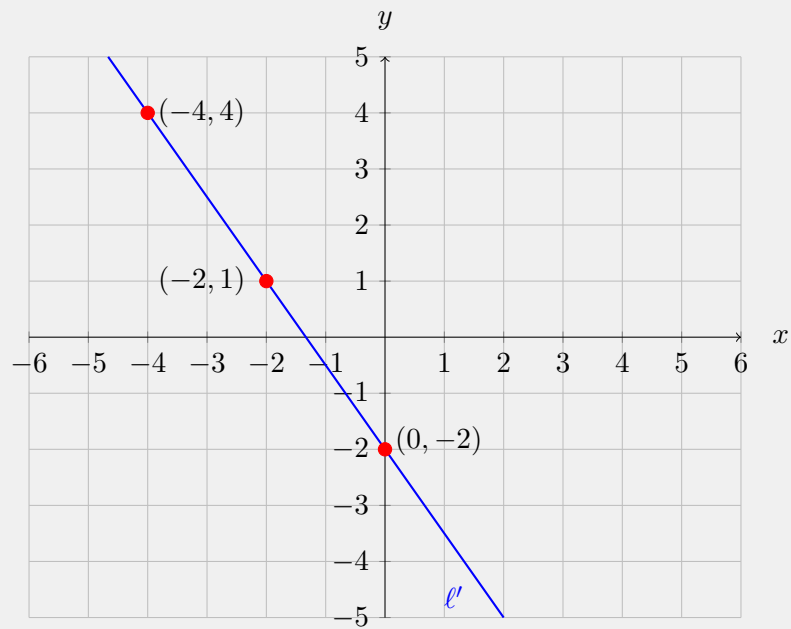
$$\vec{v}_0 + \vec{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix},$$

and

$$\vec{v}_0 - \vec{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ -3 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}.$$

The affine line passes through these three points and is parallel to the line drawn in Activity 2. Its equation is

$$3x + 2y = -4.$$



Activity 4: A Line and Its Perpendicular Line

Let ℓ be the line drawn in Activity 2:

$$\ell = \text{span} \left(\begin{bmatrix} 2 \\ -3 \end{bmatrix} \right).$$

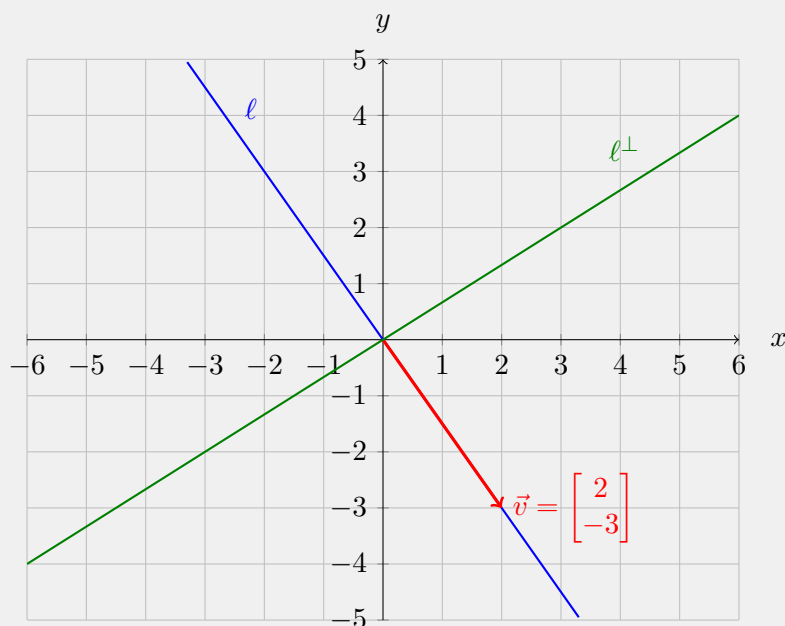
- a) i. Recall that $\ell^\perp$ is the line through the origin that is perpendicular to ℓ . Draw $\ell^\perp$ on the coordinate plane from Activity 2.
- ii. Find a nonzero vector $\vec{w}$ on $\ell^\perp$.

$$\vec{w} = \begin{bmatrix} \quad \\ \quad \end{bmatrix}.$$

- iii. Check that $\vec{w}$ is perpendicular to $\vec{v}$:

$$\vec{w} \cdot \vec{v} = \begin{bmatrix} \quad \\ \quad \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -3 \end{bmatrix} =$$

Solution: For (i), drawing $\ell^\perp$ on top of the graph from Activity 2 gives us:



For (ii), one possible choice is

$$\vec{w} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Indeed, we check at (iii) that

$$\vec{w} \cdot \vec{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ -3 \end{bmatrix} = 6 - 6 = 0.$$

Therefore,

$$\ell^\perp = \text{span} \left(\begin{bmatrix} 3 \\ 2 \end{bmatrix} \right).$$

Thus, $\ell^\perp$ is the line through the origin in the direction

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Any nonzero scalar multiple of

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

is also a valid choice for $\vec{w}$.

b) Use $\vec{w}$ and the dot product to write an equation for ℓ .

i. Equation in dot-product form:

$$\vec{w} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \underline{\hspace{2cm}}.$$

ii. Equation written explicitly in x, y coordinates:

Solution:

i. Using

$$\vec{w} = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

the equation in dot-product form is

$$\vec{w} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

Equivalently,

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

ii. Written explicitly in coordinates, this is

$$3x + 2y = 0.$$

c) Use $\vec{v}$ and the dot product to write an equation for $\ell^\perp$.

i. Equation in dot-product form:

$$\vec{v} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \underline{\hspace{2cm}}.$$

ii. Equation written explicitly in x, y coordinates:

Solution:

- i. Since $\ell^\perp$ consists of the vectors perpendicular to $\vec{v}$, its equation in dot-product form is

$$\vec{v} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

Equivalently,

$$\begin{bmatrix} 2 \\ -3 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0.$$

- ii. Written explicitly in coordinates, this is

$$2x - 3y = 0.$$

Activity 5: An Affine Line in Equation Form

Let ℓ' be the affine line drawn in Activity 3:

$$\ell' = \vec{v}_0 + \text{span}(\vec{v}),$$

where

$$\vec{v}_0 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Use $\vec{v}_0$, the vector $\vec{w}$ from Activity 4, and the dot product to write an equation for ℓ' .

a) First compute the constant on the right-hand side:

$$\vec{w} \cdot \vec{v}_0 = \begin{bmatrix} \quad \\ \quad \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \underline{\hspace{2cm}}.$$

Solution: Using

$$\vec{w} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \text{and} \quad \vec{v}_0 = \begin{bmatrix} -2 \\ 1 \end{bmatrix},$$

we obtain

$$\begin{aligned} \vec{w} \cdot \vec{v}_0 &= \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix} \\ &= 3(-2) + 2(1) \\ &= -6 + 2 \\ &= -4. \end{aligned}$$

b) i. Find the equation in dot-product form:

$$\vec{w} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \vec{w} \cdot \vec{v}_0 = \underline{\hspace{2cm}}.$$

ii. Equation written explicitly in x, y coordinates:

Solution: (i) From the previous subactivity,

$$\vec{w} \cdot \vec{v}_0 = -4.$$

Therefore, the equation in dot-product form is

$$\vec{w} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = -4.$$

(ii) Using

$$\vec{w} = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

this becomes

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = -4.$$

Written explicitly in coordinates, the equation is

$$3x + 2y = -4.$$

Activity 6: Orthonormal Bases

Which of the following pairs of vectors are orthonormal bases of $\mathbb{R}^2$?

For each pair, compute the length of each vector and the dot product of the two vectors. Explain your answer.

a)

$$\vec{u} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}.$$

$$\|\vec{u}\| = \underline{\hspace{2cm}}, \quad \|\vec{v}\| = \underline{\hspace{2cm}}, \quad \vec{u} \cdot \vec{v} = \underline{\hspace{2cm}}.$$

Is $(\vec{u}, \vec{v})$ an orthonormal basis? ☐ Yes ☐ No

Solution: ☒ Yes ☐ No

For $\vec{u}$ and $\vec{v}$ to form an orthonormal basis of $\mathbb{R}^2$, both vectors must have length 1 and be perpendicular to each other (their dot product must be 0).

$$\|\vec{u}\| = \sqrt{\frac{1}{5} + \frac{4}{5}} = 1, \quad \|\vec{v}\| = \sqrt{\frac{4}{5} + \frac{1}{5}} = 1,$$

and

$$\vec{u} \cdot \vec{v} = -\frac{2}{5} + \frac{2}{5} = 0.$$

Therefore this pair is an orthonormal basis.

b)

$$\vec{u} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} \frac{4}{5} \\ \frac{3}{5} \end{bmatrix}.$$

$$\|\vec{u}\| = \underline{\hspace{2cm}}, \quad \|\vec{v}\| = \underline{\hspace{2cm}}, \quad \vec{u} \cdot \vec{v} = \underline{\hspace{2cm}}.$$

Is $(\vec{u}, \vec{v})$ an orthonormal basis? ☐ Yes ☐ No

Solution: ☐ Yes ☒ No

$$\|\vec{u}\| = 1, \quad \|\vec{v}\| = 1,$$

but

$$\vec{u} \cdot \vec{v} = \frac{12}{25} + \frac{12}{25} = \frac{24}{25} \neq 0.$$

Both vectors are unit vectors, but they are not orthogonal. Therefore, this pair is not an orthonormal basis.

c)

$$\vec{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

$$\|\vec{u}\| = \underline{\hspace{2cm}}, \quad \|\vec{v}\| = \underline{\hspace{2cm}}, \quad \vec{u} \cdot \vec{v} = \underline{\hspace{2cm}}.$$

Is $(\vec{u}, \vec{v})$ an orthonormal basis? ☐ Yes ☐ No

Solution: ☐ Yes ☒ No

$$\|\vec{u}\| = \sqrt{5}, \quad \|\vec{v}\| = \sqrt{5},$$

and

$$\vec{u} \cdot \vec{v} = 2 - 2 = 0.$$

The vectors are orthogonal, but they are not unit vectors. Therefore, this pair is not an orthonormal basis.

Activity 7: Coordinates in an Orthonormal Basis

Recall, only one of the three pairs $(\vec{u}, \vec{v})$ from Activity 6 are an orthonormal basis of $\mathbb{R}^2$. Write those two vectors here again for your convenience.

$$\vec{u} = \begin{bmatrix} \\ \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} \\ \end{bmatrix}.$$

Now let

$$\vec{x} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

Our goal in this activity is to write $\vec{x}$ as a **linear combination** of $\vec{u}$ and $\vec{v}$. That is, find scalars a and b such that

$$\vec{x} = a\vec{u} + b\vec{v}.$$

Because $(\vec{u}, \vec{v})$ is an orthonormal basis, the coefficients can be found using dot products:

$$a = \vec{x} \cdot \vec{u}, \quad b = \vec{x} \cdot \vec{v}.$$

Fill in the blanks below.

Compute a :

$$\begin{aligned} a &= \vec{x} \cdot \vec{u} \\ &= \underline{\hspace{2cm}}. \end{aligned}$$

Compute b :

$$\begin{aligned} b &= \vec{x} \cdot \vec{v} \\ &= \underline{\hspace{2cm}}. \end{aligned}$$

Therefore,

$$\vec{x} = \underline{\hspace{2cm}} \vec{u} + \underline{\hspace{2cm}} \vec{v}.$$

Solution: The orthonormal basis from the previous activity is

$$\vec{u} = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix}.$$

Since the basis is orthonormal, the coefficients are given by the dot products of $\vec{x}$ with the basis vectors.

First,

$$\begin{aligned} a &= \vec{x} \cdot \vec{u} \\ &= \begin{bmatrix} 4 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} \\ &= \frac{4}{\sqrt{5}} - \frac{2}{\sqrt{5}} \\ &= \frac{2}{\sqrt{5}}. \end{aligned}$$

Next,

$$\begin{aligned} b &= \vec{x} \cdot \vec{v} \\ &= \begin{bmatrix} 4 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \\ &= -\frac{8}{\sqrt{5}} - \frac{1}{\sqrt{5}} \\ &= -\frac{9}{\sqrt{5}}. \end{aligned}$$

Therefore,

$$\boxed{\vec{x} = \frac{2}{\sqrt{5}} \vec{u} - \frac{9}{\sqrt{5}} \vec{v}.}$$

We can check the answer:

$$\begin{aligned} \frac{2}{\sqrt{5}} \vec{u} - \frac{9}{\sqrt{5}} \vec{v} &= \frac{2}{\sqrt{5}} \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \end{bmatrix} - \frac{9}{\sqrt{5}} \begin{bmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{5} \\ \frac{4}{5} \end{bmatrix} + \begin{bmatrix} \frac{18}{5} \\ -\frac{9}{5} \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ -1 \end{bmatrix} \\ &= \vec{x}. \end{aligned}$$